Assessment 7

Course: Database Systems Lab
Course Code: BCSE302L
Slot : L39 + L40

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Create the restaurants collection in your DB [24BYB0117]
Import the json file to your collection [restaurants.json]

```
roxxon2@roxxon:~$ sudo docker cp ~/restaurants.json college-mongodb:/tmp/restaurants.json
Successfully copied 2.03MB (transferred 2.03MB) to college-mongodb:/tmp/restaurants.json
roxxon2@roxxon:~$ sudo docker exec -it college-mongodb mongoimport \
  --host 127.0.0.1 \
  --port 27017 \
  -u shubham \
  -p 'hello123' \
  --authenticationDatabase admin \
  --db 24BYB0117 \
  --collection restaurants \
  --file /tmp/restaurants.json
2026-10-23T13:22:18.797+0000    connected to: mongodb://127.0.0.1:27017/
2026-10-23T13:22:19.046+0000    3772 document(s) imported successfully. 0 document(s) failed to import.
roxxon2@roxxon:~$ sudo docker exec -it college-mongodb mongosh \
  -u shubham -p 'hello123' \
  --authenticationDatabase admin
```

```
test> use 24BYB0117
switched to db 24BYB0117
24BYB0117> db.restaurants.countDocuments()
3772
24BYB0117>
```

1. Display all the documents in the collection restaurants.

```
24BYB0117> // Q1. Display all the documents in the collection restaurants.
| db.restaurants.find({});
[
  {
    _id: ObjectId('6adb5f8adeafff9f132118b9'),
    address: {
      building: '1007',
      coord: [ -73.856077, 40.848447 ],
      street: 'Morris Park Ave',
      zipcode: '10462'
    },
    borough: 'Bronx',
    cuisine: 'Bakery',
    grades: [
      { date: ISODate('2014-03-03T00:00:00.000Z'), grade: 'A', score: 2 },
      { date: ISODate('2013-09-11T00:00:00.000Z'), grade: 'A', score: 6 },
      { date: ISODate('2013-01-24T00:00:00.000Z'), grade: 'A', score: 10 },
      { date: ISODate('2011-11-23T00:00:00.000Z'), grade: 'A', score: 9 },
      { date: ISODate('2011-03-10T00:00:00.000Z'), grade: 'B', score: 14 }
    ],
    name: 'Morris Park Bake Shop',
    restaurant_id: '30075445'
  },
  {
    _id: ObjectId('6adb5f8adeafff9f132118ba'),
    address: {
      building: '469',
      coord: [ -73.961704, 40.662942 ],
      street: 'Flatbush Avenue',
      zipcode: '11225'
    },
    borough: 'Brooklyn',
    cuisine: 'Hamburgers',
    grades: [
      { date: ISODate('2014-12-30T00:00:00.000Z'), grade: 'A', score: 8 },
      { date: ISODate('2014-07-01T00:00:00.000Z'), grade: 'B', score: 23 },
      { date: ISODate('2013-04-30T00:00:00.000Z'), grade: 'A', score: 12 },
      { date: ISODate('2012-05-08T00:00:00.000Z'), grade: 'A', score: 12 }
    ],
    name: "Wendy'S",
    restaurant_id: '30112340'
  },
  {
    _id: ObjectId('6adb5f8adeafff9f132118bb'),
    address: {
      building: '2780',
      coord: [ -73.98241999999999, 40.579505 ],
      street: 'Stillwell Avenue',
      zipcode: '11224'
    },
    borough: 'Brooklyn',
    cuisine: 'American ',
    grades: [
      { date: ISODate('2014-06-10T00:00:00.000Z'), grade: 'A', score: 5 },
      { date: ISODate('2013-06-05T00:00:00.000Z'), grade: 'A', score: 7 },
      { date: ISODate('2012-04-13T00:00:00.000Z'), grade: 'A', score: 12 },
      { date: ISODate('2011-10-12T00:00:00.000Z'), grade: 'A', score: 12 }
    ],
    name: 'Riviera Caterer',
    restaurant_id: '40356018'
  },
```

2. Display the fields restaurant_id, name, borough and cuisine for all the documents in the collection restaurant.

```
24BYB0117> // Q2. Display the fields restaurant_id, name, borough and cuisine for all the documents in the collection restaurant.
| db.restaurants.find(
|   {},
|   { _id: 0, restaurant_id: 1, name: 1, borough: 1, cuisine: 1 }
| );
[
  {
    borough: 'Bronx',
    cuisine: 'Bakery',
    name: 'Morris Park Bake Shop',
    restaurant_id: '30075445'
  },
  {
    borough: 'Brooklyn',
    cuisine: 'Hamburgers',
    name: "Wendy'S",
    restaurant_id: '30112340'
  },
  {
    borough: 'Brooklyn',
    cuisine: 'American ',
    name: 'Riviera Caterer',
    restaurant_id: '40356018'
  },
  {
    borough: 'Manhattan',
    cuisine: 'Irish',
    name: 'Dj Reynolds Pub And Restaurant',
    restaurant_id: '30191841'
  },
```

```
  {
    borough: 'Queens',
    cuisine: 'American ',
    name: 'Brunos On The Boulevard',
    restaurant_id: '40356151'
  },
  {
    borough: 'Brooklyn',
    cuisine: 'American ',
    name: 'Regina Caterers',
    restaurant_id: '40356649'
  },
  {
    borough: 'Brooklyn',
    cuisine: 'Delicatessen',
    name: "Wilken'S Fine Food",
    restaurant_id: '40356483'
  },
  {
    borough: 'Staten Island',
    cuisine: 'Jewish/Kosher',
    name: 'Kosher Island',
    restaurant_id: '40356442'
  },
  {
    borough: 'Brooklyn',
    cuisine: 'Ice Cream, Gelato, Yogurt, Ices',
    name: 'Taste The Tropics Ice Cream',
    restaurant_id: '40356731'
  },
```

3. Display all the restaurant which is in the borough Bronx.

```
24BYB0117> // Q3. Display all the restaurant which is in the borough Bronx.
| db.restaurants.find({ borough: "Bronx" });
[
  {
    _id: ObjectId('6adb5f8adeafff9f132118b9'),
    address: {
      building: '1007',
      coord: [ -73.856077, 40.848447 ],
      street: 'Morris Park Ave',
      zipcode: '10462'
    },
    borough: 'Bronx',
    cuisine: 'Bakery',
    grades: [
      { date: ISODate('2014-03-03T00:00:00.000Z'), grade: 'A', score: 2 },
      { date: ISODate('2013-09-11T00:00:00.000Z'), grade: 'A', score: 6 },
      { date: ISODate('2013-01-24T00:00:00.000Z'), grade: 'A', score: 10 },
      { date: ISODate('2011-11-23T00:00:00.000Z'), grade: 'A', score: 9 },
      { date: ISODate('2011-03-10T00:00:00.000Z'), grade: 'B', score: 14 }
    ],
    name: 'Morris Park Bake Shop',
    restaurant_id: '30075445'
  },
  {
    _id: ObjectId('6adb5f8adeafff9f132118c2'),
    address: {
      building: '2300',
      coord: [ -73.8786113, 40.8502883 ],
      street: 'Southern Boulevard',
      zipcode: '10460'
```

```
    },
    borough: 'Bronx',
    cuisine: 'American ',
    grades: [
      { date: ISODate('2014-05-28T00:00:00.000Z'), grade: 'A', score: 11 },
      { date: ISODate('2013-06-19T00:00:00.000Z'), grade: 'A', score: 4 },
      { date: ISODate('2012-06-15T00:00:00.000Z'), grade: 'A', score: 3 }
    ],
    name: 'Wild Asia',
    restaurant_id: '40357217'
  },
  {
    _id: ObjectId('6adb5f8adeafff9f132118d7'),
    address: {
      building: '1006',
      coord: [ -73.84856870000002, 40.8903781 ],
      street: 'East 233 Street',
      zipcode: '10466'
    },
    borough: 'Bronx',
    cuisine: 'Ice Cream, Gelato, Yogurt, Ices',
    grades: [
      { date: ISODate('2014-04-24T00:00:00.000Z'), grade: 'A', score: 10 },
      { date: ISODate('2013-09-05T00:00:00.000Z'), grade: 'A', score: 10 },
      { date: ISODate('2013-02-21T00:00:00.000Z'), grade: 'A', score: 9 },
      { date: ISODate('2012-07-03T00:00:00.000Z'), grade: 'A', score: 11 },
      { date: ISODate('2011-07-11T00:00:00.000Z'), grade: 'A', score: 5 }
    ],
```

4. Display the next 5 restaurants after skipping first 5 which are in the borough Bronx.

```
24BYB0117> // Q4. Display the next 5 restaurants after skipping first 5 which are in the borough Bronx.
| db.restaurants.find({ borough: "Bronx" }).skip(5).limit(5);
[
  {
    _id: ObjectId('6adb5f8adeafff9f132118f5'),
    address: {
      building: '658',
      coord: [ -73.81363999999999, 40.82941100000001 ],
      street: 'Clarence Ave',
      zipcode: '10465'
    },
    borough: 'Bronx',
    cuisine: 'American ',
    grades: [
      { date: ISODate('2014-06-21T00:00:00.000Z'), grade: 'A', score: 5 },
      { date: ISODate('2012-07-11T00:00:00.000Z'), grade: 'A', score: 10 }
    ],
    name: 'Manhem Club',
    restaurant_id: '40364363'
  },
  {
    _id: ObjectId('6adb5f8adeafff9f1321190d'),
    address: {
      building: '2222',
      coord: [ -73.84971759999999, 40.8304811 ],
      street: 'Haviland Avenue',
      zipcode: '10462'
    },
    borough: 'Bronx',
    cuisine: 'American ',
    grades: [
      { date: ISODate('2014-12-18T00:00:00.000Z'), grade: 'A', score: 7 },
      { date: ISODate('2014-05-01T00:00:00.000Z'), grade: 'B', score: 17 },
      { date: ISODate('2013-03-14T00:00:00.000Z'), grade: 'A', score: 12 },
      { date: ISODate('2012-09-20T00:00:00.000Z'), grade: 'A', score: 9 },
      { date: ISODate('2012-02-08T00:00:00.000Z'), grade: 'B', score: 19 }
```

```
    name: 'The New Starling Athletic Club Of The Bronx',
    restaurant_id: '40364956'
  },
  {
    _id: ObjectId('6adb5f8adeafff9f13211927'),
    address: {
      building: '72',
      coord: [ -73.92506, 40.8275556 ],
      street: 'East  161 Street',
      zipcode: '10451'
    },
    borough: 'Bronx',
    cuisine: 'American ',
    grades: [
      { date: ISODate('2014-04-15T00:00:00.000Z'), grade: 'A', score: 9 },
      { date: ISODate('2013-11-14T00:00:00.000Z'), grade: 'A', score: 4 },
      { date: ISODate('2013-07-29T00:00:00.000Z'), grade: 'A', score: 10 },
      { date: ISODate('2012-12-31T00:00:00.000Z'), grade: 'B', score: 15 },
      { date: ISODate('2012-05-30T00:00:00.000Z'), grade: 'A', score: 13 },
      { date: ISODate('2012-01-09T00:00:00.000Z'), grade: 'A', score: 10 },
      { date: ISODate('2011-08-15T00:00:00.000Z'), grade: 'C', score: 37 }
    ],
    name: 'Yankee Tavern',
    restaurant_id: '40365499'
  },
  {
    _id: ObjectId('6adb5f8adeafff9f13211939'),
    address: {
      building: '331',
      coord: [ -73.87786539999999, 40.8724377 ],
      street: 'East  204 Street',
      zipcode: '10467'
    },
    borough: 'Bronx',
    cuisine: 'Irish',
```

```
    borough: 'Bronx',
    cuisine: 'Irish',
    grades: [
      { date: ISODate('2014-08-26T00:00:00.000Z'), grade: 'A', score: 10 },
      { date: ISODate('2014-03-26T00:00:00.000Z'), grade: 'B', score: 23 },
      { date: ISODate('2013-09-11T00:00:00.000Z'), grade: 'A', score: 13 },
      { date: ISODate('2012-12-18T00:00:00.000Z'), grade: 'B', score: 27 },
      { date: ISODate('2011-10-20T00:00:00.000Z'), grade: 'A', score: 13 }
    ],
    name: 'Mcdwyers Pub',
    restaurant_id: '40365893'
  },
  {
    _id: ObjectId('6adb5f8adeafff9f13211951'),
    address: {
      building: '5820',
      coord: [ -73.9002615, 40.885186 ],
      street: 'Broadway',
      zipcode: '10463'
    },
    borough: 'Bronx',
    cuisine: 'American ',
    grades: [
      { date: ISODate('2014-02-26T00:00:00.000Z'), grade: 'A', score: 5 },
      { date: ISODate('2013-10-09T00:00:00.000Z'), grade: 'B', score: 19 },
      { date: ISODate('2013-05-15T00:00:00.000Z'), grade: 'A', score: 9 },
      { date: ISODate('2012-11-20T00:00:00.000Z'), grade: 'B', score: 18 },
      { date: ISODate('2011-10-17T00:00:00.000Z'), grade: 'A', score: 10 },
      { date: ISODate('2011-06-22T00:00:00.000Z'), grade: 'C', score: 35 }
    ],
    name: 'The Punch Bowl',
    restaurant_id: '40366497'
  }
]
24BYB0117>
```

5. Find the restaurants who achieved a score more than 90.

```
24BYB0117> // Q4. Display the next 5 restaurants after skipping first 5 which are in the borough Bronx.
| db.restaurants.find({ borough: "Bronx" }).skip(5).limit(5);
[
  {
    _id: ObjectId('6adb5f8adeafff9f132118f5'),
    address: {
      building: '658',
      coord: [ -73.81363999999999, 40.82941100000001 ],
      street: 'Clarence Ave',
      zipcode: '10465'
    },
    borough: 'Bronx',
    cuisine: 'American ',
    grades: [
      { date: ISODate('2014-06-21T00:00:00.000Z'), grade: 'A', score: 5 },
      { date: ISODate('2012-07-11T00:00:00.000Z'), grade: 'A', score: 10 }
    ],
    name: 'Manhem Club',
    restaurant_id: '40364363'
  },
  {
    _id: ObjectId('6adb5f8adeafff9f1321190d'),
    address: {
      building: '2222',
      coord: [ -73.84971759999999, 40.8304811 ],
      street: 'Haviland Avenue',
      zipcode: '10462'
    },
    borough: 'Bronx',
    cuisine: 'American ',
    grades: [
      { date: ISODate('2014-12-18T00:00:00.000Z'), grade: 'A', score: 7 },
      { date: ISODate('2014-05-01T00:00:00.000Z'), grade: 'B', score: 17 },
      { date: ISODate('2013-03-14T00:00:00.000Z'), grade: 'A', score: 12 },
      { date: ISODate('2012-09-20T00:00:00.000Z'), grade: 'A', score: 9 },
```

```
    name: 'The New Starling Athletic Club Of The Bronx',
    restaurant_id: '40364956'
  },
  {
    _id: ObjectId('6adb5f8adeafff9f13211927'),
    address: {
      building: '72',
      coord: [ -73.92506, 40.8275556 ],
      street: 'East  161 Street',
      zipcode: '10451'
    },
    borough: 'Bronx',
    cuisine: 'American ',
    grades: [
      { date: ISODate('2014-04-15T00:00:00.000Z'), grade: 'A', score: 9 },
      { date: ISODate('2013-11-14T00:00:00.000Z'), grade: 'A', score: 4 },
      { date: ISODate('2013-07-29T00:00:00.000Z'), grade: 'A', score: 10 },
      { date: ISODate('2012-12-31T00:00:00.000Z'), grade: 'B', score: 15 },
      { date: ISODate('2012-05-30T00:00:00.000Z'), grade: 'A', score: 13 },
      { date: ISODate('2012-01-09T00:00:00.000Z'), grade: 'A', score: 10 },
      { date: ISODate('2011-08-15T00:00:00.000Z'), grade: 'C', score: 37 }
    ],
    name: 'Yankee Tavern',
    restaurant_id: '40365499'
  },
  {
    _id: ObjectId('6adb5f8adeafff9f13211939'),
    address: {
      building: '331',
      coord: [ -73.87786539999999, 40.8724377 ],
      street: 'East  204 Street',
      zipcode: '10467'
    },
    borough: 'Bronx',
    cuisine: 'Irish',
    grades: [
      { date: ISODate('2014-08-26T00:00:00.000Z'), grade: 'A', score: 10 },
```

```
    borough: 'Bronx',
    cuisine: 'Irish',
    grades: [
      { date: ISODate('2014-08-26T00:00:00.000Z'), grade: 'A', score: 10 },
      { date: ISODate('2014-03-26T00:00:00.000Z'), grade: 'B', score: 23 },
      { date: ISODate('2013-09-11T00:00:00.000Z'), grade: 'A', score: 13 },
      { date: ISODate('2012-12-18T00:00:00.000Z'), grade: 'B', score: 27 },
      { date: ISODate('2011-10-20T00:00:00.000Z'), grade: 'A', score: 13 }
    ],
    name: 'Mcdwyers Pub',
    restaurant_id: '40365893'
  },
  {
    _id: ObjectId('6adb5f8adeafff9f13211951'),
    address: {
      building: '5820',
      coord: [ -73.9002615, 40.885186 ],
      street: 'Broadway',
      zipcode: '10463'
    },
    borough: 'Bronx',
    cuisine: 'American ',
    grades: [
      { date: ISODate('2014-02-26T00:00:00.000Z'), grade: 'A', score: 5 },
      { date: ISODate('2013-10-09T00:00:00.000Z'), grade: 'B', score: 19 },
      { date: ISODate('2013-05-15T00:00:00.000Z'), grade: 'A', score: 9 },
      { date: ISODate('2012-11-20T00:00:00.000Z'), grade: 'B', score: 18 },
      { date: ISODate('2011-10-17T00:00:00.000Z'), grade: 'A', score: 10 },
      { date: ISODate('2011-06-22T00:00:00.000Z'), grade: 'C', score: 35 }
    ],
    name: 'The Punch Bowl',
    restaurant_id: '40366497'
  }
]
```

6. Find the restaurants that do not prepare any cuisine of 'American' and their grade score more than 70 and latitude less than -65.754168.

```
24BYB0117> // Q6. Find the restaurants that do not prepare any cuisine of 'American' and their grade score more than 70 and latitude less than -65.754168
| db.restaurants.find({
|   cuisine: { $nin: ["American", "American "] },
|   "grades.score": { $gt: 70 },
|   "address.coord.0": { $lt: -65.754168 }
| });
[
  {
    _id: ObjectId('6adb5f8adeafff9f13211ab9'),
    address: {
      building: '345',
      coord: [ -73.9864626, 40.7266739 ],
      street: 'East 6 Street',
      zipcode: '10003'
    },
    borough: 'Manhattan',
    cuisine: 'Indian',
    grades: [
      { date: ISODate('2014-09-15T00:00:00.000Z'), grade: 'A', score: 5 },
      { date: ISODate('2014-01-14T00:00:00.000Z'), grade: 'A', score: 8 },
      { date: ISODate('2013-05-30T00:00:00.000Z'), grade: 'A', score: 12 },
      { date: ISODate('2013-04-24T00:00:00.000Z'), grade: 'P', score: 2 },
      { date: ISODate('2012-10-01T00:00:00.000Z'), grade: 'A', score: 9 },
      { date: ISODate('2012-04-06T00:00:00.000Z'), grade: 'C', score: 92 },
      { date: ISODate('2011-11-03T00:00:00.000Z'), grade: 'C', score: 41 }
    ],
    name: 'Gandhi',
    restaurant_id: '40381295'
  },
  {
    _id: ObjectId('6adb5f8adeafff9f13211c1d'),
    address: {
      building: '130',
      coord: [ -73.984758, 40.7457939 ],
      street: 'Madison Avenue',
      zipcode: '10016'
    },
    borough: 'Manhattan',
    cuisine: 'Pizza/Italian',
    grades: [
      { date: ISODate('2014-12-24T00:00:00.000Z'), grade: 'Z', score: 31 },
      { date: ISODate('2014-06-17T00:00:00.000Z'), grade: 'C', score: 98 },
      { date: ISODate('2013-12-12T00:00:00.000Z'), grade: 'C', score: 32 },
      { date: ISODate('2013-05-22T00:00:00.000Z'), grade: 'B', score: 21 },
      { date: ISODate('2012-05-02T00:00:00.000Z'), grade: 'A', score: 11 }
    ],
    name: 'Bella Napoli',
    restaurant_id: '40393488'
  },
  {
    _id: ObjectId('6adb5f8adeafff9f13211c26'),
    address: {
      building: '101',
      coord: [ -73.9243061, 40.8276297 ],
      street: 'East 161 Street',
      zipcode: '10451'
    },
    borough: 'Bronx',
    cuisine: 'Latin (Cuban, Dominican, Puerto Rican, South & Central American)',
    grades: [
      { date: ISODate('2014-04-10T00:00:00.000Z'), grade: 'A', score: 10 },
      { date: ISODate('2013-10-01T00:00:00.000Z'), grade: 'A', score: 6 },
      { date: ISODate('2013-04-11T00:00:00.000Z'), grade: 'B', score: 25 },
      { date: ISODate('2012-10-25T00:00:00.000Z'), grade: 'A', score: 12 },
      { date: ISODate('2012-10-10T00:00:00.000Z'), grade: 'P', score: 12 },
      { date: ISODate('2012-05-25T00:00:00.000Z'), grade: 'B', score: 14 },
      { date: ISODate('2011-09-14T00:00:00.000Z'), grade: 'B', score: 26 },
      { date: ISODate('2011-04-25T00:00:00.000Z'), grade: 'C', score: 76 }
    ],
    name: 'El Molino Rojo Restaurant',
    restaurant_id: '40393688'
```

```
    coord: [ -73.94610279999999, 40.7137587 ],
    street: 'Manhattan Avenue',
    zipcode: '11211'
  },
  borough: 'Brooklyn',
  cuisine: 'Bakery',
  grades: [
    { date: ISODate('2014-03-19T00:00:00.000Z'), grade: 'A', score: 10 },
    { date: ISODate('2013-10-10T00:00:00.000Z'), grade: 'A', score: 10 },
    { date: ISODate('2013-05-07T00:00:00.000Z'), grade: 'A', score: 13 },
    { date: ISODate('2012-09-11T00:00:00.000Z'), grade: 'B', score: 18 },
    { date: ISODate('2012-04-18T00:00:00.000Z'), grade: 'A', score: 9 },
    { date: ISODate('2011-09-27T00:00:00.000Z'), grade: 'A', score: 8 },
    { date: ISODate('2011-05-03T00:00:00.000Z'), grade: 'C', score: 77 }
  ],
  name: 'Fortunato Bros Cafe & Bakery',
  restaurant_id: '40400561'
},
{
  _id: ObjectId('6adb5f8adeafff9f132123a0'),
  address: {
    building: '231',
    coord: [ -73.9772294, 40.7527262 ],
    street: 'Grand Central Station',
    zipcode: '10017'
  },
  borough: 'Manhattan',
  cuisine: 'Italian',
  grades: [
    { date: ISODate('2015-01-07T00:00:00.000Z'), grade: 'Z', score: 20 },
    { date: ISODate('2014-07-03T00:00:00.000Z'), grade: 'B', score: 14 },
    { date: ISODate('2013-12-21T00:00:00.000Z'), grade: 'A', score: 13 },
    { date: ISODate('2013-05-17T00:00:00.000Z'), grade: 'C', score: 76 },
    { date: ISODate('2012-04-20T00:00:00.000Z'), grade: 'A', score: 12 }
  ],
  name: 'Two Boots Grand Central',
  restaurant_id: '40725591'
```

7. Find the restaurant Id, name, borough and cuisine for those restaurants which contain 'Wil' as first three letters for its name.

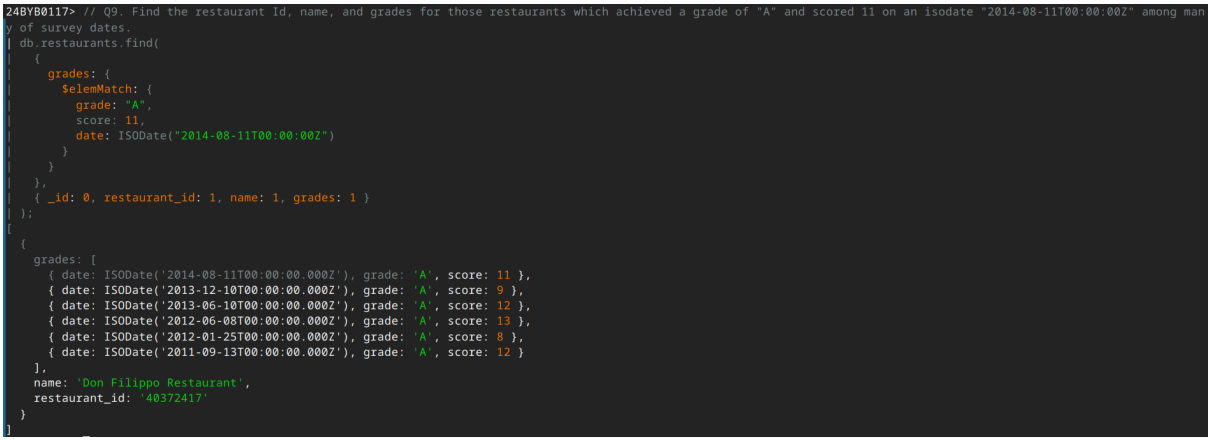
```
24BYB0117> // Q7. Find the restaurant Id, name, borough and cuisine for those restaurants which contain 'Wil' as first three letters for its name.
| db.restaurants.find(
|   { name: /^Wil/ },
|   { _id: 0, restaurant_id: 1, name: 1, borough: 1, cuisine: 1 }
| );
|
[
  {
    borough: 'Brooklyn',
    cuisine: 'Delicatessen',
    name: "Wilken'S Fine Food",
    restaurant_id: '40356483'
  },
  {
    borough: 'Bronx',
    cuisine: 'American ',
    name: 'Wild Asia',
    restaurant_id: '40357217'
  },
  {
    borough: 'Bronx',
    cuisine: 'Pizza',
    name: 'Wilbel Pizza',
    restaurant_id: '40871979'
  }
]
```

8. Find the restaurant Id, name, borough and cuisine for those restaurants which belong to the borough Staten Island or Queens or Bronx or Brooklyn.

```
24BYB0117> // Q8. Find the restaurant Id, name, borough and cuisine for those restaurants which belong to the borough Staten Island or Queens or Bronx or Brooklyn.
| db.restaurants.find(
|   { borough: { $in: ["Staten Island", "Queens", "Bronx", "Brooklyn"] } },
|   { _id: 0, restaurant_id: 1, name: 1, borough: 1, cuisine: 1 }
| );
[
  {
    borough: 'Bronx',
    cuisine: 'Bakery',
    name: 'Morris Park Bake Shop',
    restaurant_id: '30075445'
  },
  {
    borough: 'Brooklyn',
    cuisine: 'Hamburgers',
    name: "Wendy'S",
    restaurant_id: '30112340'
  },
  {
    borough: 'Brooklyn',
    cuisine: 'American ',
    name: 'Riviera Caterer',
    restaurant_id: '40356018'
  },
  {
    borough: 'Queens',
    cuisine: 'American ',
    name: 'Brunos On The Boulevard',
    restaurant_id: '40356151'
  },
  {
    borough: 'Brooklyn',
    cuisine: 'American ',
    name: 'Regina Caterers',
    restaurant_id: '40356649'
  },
```

```
{
  borough: 'Brooklyn',
  cuisine: 'Delicatessen',
  name: "Wilken'S Fine Food",
  restaurant_id: '40356483'
},
{
  borough: 'Staten Island',
  cuisine: 'Jewish/Kosher',
  name: 'Kosher Island',
  restaurant_id: '40356442'
},
{
  borough: 'Brooklyn',
  cuisine: 'Ice Cream, Gelato, Yogurt, Ices',
  name: 'Taste The Tropics Ice Cream',
  restaurant_id: '40356731'
},
{
  borough: 'Bronx',
  cuisine: 'American ',
  name: 'Wild Asia',
  restaurant_id: '40357217'
},
{
  borough: 'Brooklyn',
  cuisine: 'American ',
  name: 'C & C Catering Service',
  restaurant_id: '40357437'
},
{
  borough: 'Brooklyn',
  cuisine: 'Chinese',
  name: 'May May Kitchen',
  restaurant_id: '40358429'
},
```

9. Find the restaurant Id, name, and grades for those restaurants which achieved a grade of "A" and scored 11 on an isodate "2014-08-11T00:00:00Z" among many survey dates.



```
24BYB0117> // Q9. Find the restaurant Id, name, and grades for those restaurants which achieved a grade of "A" and scored 11 on an isodate "2014-08-11T00:00:00Z" among many of survey dates.
| db.restaurants.find(
|   {
|     grades: {
|       $elemMatch: {
|         grade: "A",
|         score: 11,
|         date: ISODate("2014-08-11T00:00:00Z")
|       }
|     }
|   },
|   { _id: 0, restaurant_id: 1, name: 1, grades: 1 }
| );
[
  {
    grades: [
      { date: ISODate('2014-08-11T00:00:00.000Z'), grade: 'A', score: 11 },
      { date: ISODate('2013-12-10T00:00:00.000Z'), grade: 'A', score: 9 },
      { date: ISODate('2013-06-10T00:00:00.000Z'), grade: 'A', score: 12 },
      { date: ISODate('2012-06-08T00:00:00.000Z'), grade: 'A', score: 13 },
      { date: ISODate('2012-01-25T00:00:00.000Z'), grade: 'A', score: 8 },
      { date: ISODate('2011-09-13T00:00:00.000Z'), grade: 'A', score: 12 }
    ],
    name: 'Don Filippo Restaurant',
    restaurant_id: '40372417'
  }
]
```

10. Arrange the name of the restaurants in descending along with all the columns.

```
24BYB0117> // Q10. Arrange the name of the restaurants in descending along with all the columns
| db.restaurants.find({}).sort({ name: -1 });
[
  {
    _id: ObjectId('6adb5f8adeafff9f13211979'),
    address: {
      building: '6946',
      coord: [ -73.8811834, 40.7017759 ],
      street: 'Myrtle Avenue',
      zipcode: '11385'
    },
    borough: 'Queens',
    cuisine: 'German',
    grades: [
      { date: ISODate('2014-09-24T00:00:00.000Z'), grade: 'A', score: 11 },
      { date: ISODate('2014-04-17T00:00:00.000Z'), grade: 'A', score: 7 },
      { date: ISODate('2013-03-12T00:00:00.000Z'), grade: 'A', score: 13 },
      { date: ISODate('2012-10-02T00:00:00.000Z'), grade: 'A', score: 9 },
      { date: ISODate('2012-05-09T00:00:00.000Z'), grade: 'A', score: 13 },
      { date: ISODate('2011-12-28T00:00:00.000Z'), grade: 'B', score: 24 }
    ],
    name: 'Zum Stammtisch',
    restaurant_id: '40367377'
  },
  {
    _id: ObjectId('6adb5f8bdeafff9f132124b1'),
    address: {
      building: '107109',
      coord: [ -73.9744668, 40.731155 ],
      street: 'Avenue C',
      zipcode: '10009'
    },
    borough: 'Manhattan',
    cuisine: 'German',
    grades: [
```

```
  {
    _id: ObjectId('6adb5f8bdeafff9f132124b1'),
    address: {
      building: '107109',
      coord: [ -73.9744668, 40.731155 ],
      street: 'Avenue C',
      zipcode: '10009'
    },
    borough: 'Manhattan',
    cuisine: 'German',
    grades: [
      { date: ISODate('2014-03-04T00:00:00.000Z'), grade: 'A', score: 7 },
      { date: ISODate('2013-08-21T00:00:00.000Z'), grade: 'A', score: 13 },
      { date: ISODate('2013-02-27T00:00:00.000Z'), grade: 'A', score: 11 },
      { date: ISODate('2012-06-05T00:00:00.000Z'), grade: 'A', score: 12 },
      { date: ISODate('2011-12-20T00:00:00.000Z'), grade: 'A', score: 9 },
      { date: ISODate('2011-07-15T00:00:00.000Z'), grade: 'A', score: 6 }
    ],
    name: 'Zum Schneider',
    restaurant_id: '40763382'
  },
  {
    _id: ObjectId('6adb5f8bdeafff9f132126ec'),
    address: {
      building: '2905',
      coord: [ -73.91337299999999, 40.775064 ],
      street: '23 Avenue',
      zipcode: '11105'
    },
```

11. Find the restaurants that have at least one grade with a score of less than 5 and that are located in the borough of Manhattan.

```
24BYB0117> // Q11. Find the restaurants that have at least one grade with a score of less than 5 and that are located in the borough of Manhattan.
| db.restaurants.find({
|   borough: "Manhattan",
|   "grades.score": { $lt: 5 }
| });
[
  {
    _id: ObjectId('6adb5f8adeafff9f132118bc'),
    address: {
      building: '351',
      coord: [ -73.98513559999999, 40.7676919 ],
      street: 'West   57 Street',
      zipcode: '10019'
    },
    borough: 'Manhattan',
    cuisine: 'Irish',
    grades: [
      { date: ISODate('2014-09-06T00:00:00.000Z'), grade: 'A', score: 2 },
      { date: ISODate('2013-07-22T00:00:00.000Z'), grade: 'A', score: 11 },
      { date: ISODate('2012-07-31T00:00:00.000Z'), grade: 'A', score: 12 },
      { date: ISODate('2011-12-29T00:00:00.000Z'), grade: 'A', score: 12 }
    ],
    name: 'Dj Reynolds Pub And Restaurant',
    restaurant_id: '30191841'
  },
  {
    _id: ObjectId('6adb5f8adeafff9f132118c5'),
    address: {
      building: '1',
      coord: [ -73.96926909999999, 40.7685235 ],
      street: 'East   66 Street',
      zipcode: '10065'
    },
    borough: 'Manhattan',
    cuisine: 'American ',
    grades: [
```

12. Find the restaurants that have a grade with a score of 2 and a grade with a score of 6 and are located in the borough of Manhattan or Brooklyn, and their cuisine is not American or Chinese.

```
24BYB0117> // Q12. Find the restaurants that have a grade with a score of 2 and a grade with a score of 6 and are located in the borough of Manhattan or Brooklyn, and thei
r cuisine is not American or Chinese.
| db.restaurants.find({
|   borough: { $in: ["Manhattan", "Brooklyn"] },
|   cuisine: { $nin: ["American", "American ", "Chinese"] },
|   grades: {
|     $all: [
|       { $elemMatch: { score: 2 } },
|       { $elemMatch: { score: 6 } }
|     ]
|   }
| });
[
  {
    _id: ObjectId('6adb5f8adeafff9f132118f7'),
    address: {
      building: '261',
      coord: [ -73.94839189999999, 40.7224876 ],
      street: 'Driggs Avenue',
      zipcode: '11222'
    },
    borough: 'Brooklyn',
    cuisine: 'Polish',
    grades: [
      { date: ISODate('2014-05-31T00:00:00.000Z'), grade: 'A', score: 2 },
      { date: ISODate('2013-05-10T00:00:00.000Z'), grade: 'A', score: 3 },
      { date: ISODate('2012-02-17T00:00:00.000Z'), grade: 'A', score: 6 },
      { date: ISODate('2011-10-14T00:00:00.000Z'), grade: 'C', score: 54 }
    ],
    name: 'Polish National Home',
    restaurant_id: '40364404'
  },
  {
    _id: ObjectId('6adb5f8adeafff9f1321198b'),
    address: {
```

```
    _id: ObjectId('6adb5f8adeafff9f1321198b'),
    address: {
      building: '1900',
      coord: [ -73.9819129, 40.77151449999999 ],
      street: 'Broadway',
      zipcode: '10023'
    },
    borough: 'Manhattan',
    cuisine: 'Italian',
    grades: [
      { date: ISODate('2014-09-04T00:00:00.000Z'), grade: 'A', score: 6 },
      { date: ISODate('2014-04-01T00:00:00.000Z'), grade: 'A', score: 2 },
      { date: ISODate('2013-03-14T00:00:00.000Z'), grade: 'A', score: 7 },
      { date: ISODate('2012-09-17T00:00:00.000Z'), grade: 'A', score: 12 },
      { date: ISODate('2011-09-07T00:00:00.000Z'), grade: 'A', score: 7 }
    ],
    name: 'Fiorellos',
    restaurant_id: '40367841'
  },
  {
    _id: ObjectId('6adb5f8adeafff9f13211a26'),
    address: {
      building: '226',
      coord: [ -73.9777137, 40.6433389 ],
      street: 'Church Avenue',
      zipcode: '11218'
    },
    borough: 'Brooklyn',
    cuisine: 'Pizza/Italian',
    grades: [
      { date: ISODate('2014-09-11T00:00:00.000Z'), grade: 'A', score: 2 },
      { date: ISODate('2014-03-03T00:00:00.000Z'), grade: 'A', score: 12 },
      { date: ISODate('2012-11-29T00:00:00.000Z'), grade: 'A', score: 6 },
      { date: ISODate('2011-11-29T00:00:00.000Z'), grade: 'A', score: 8 }
    ],
```

13. Find the restaurants that have all grades with a score greater than 5 and are located in the borough of Manhattan or Brooklyn.

```
24BYB0117> // Q13. Find the restaurants that have all grades with a score greater than 5 and are located in the borough of Manhattan or Brooklyn.
| db.restaurants.find({
|   borough: { $in: ["Manhattan", "Brooklyn"] },
|   grades: { $not: { $elemMatch: { score: { $lte: 5 } } } }
| });
[
  {
    _id: ObjectId('6adb5f8adeafff9f132118ba'),
    address: {
      building: '469',
      coord: [ -73.961704, 40.662942 ],
      street: 'Flatbush Avenue',
      zipcode: '11225'
    },
    borough: 'Brooklyn',
    cuisine: 'Hamburgers',
    grades: [
      { date: ISODate('2014-12-30T00:00:00.000Z'), grade: 'A', score: 8 },
      { date: ISODate('2014-07-01T00:00:00.000Z'), grade: 'B', score: 23 },
      { date: ISODate('2013-04-30T00:00:00.000Z'), grade: 'A', score: 12 },
      { date: ISODate('2012-05-08T00:00:00.000Z'), grade: 'A', score: 12 }
    ],
    name: "Wendy'S",
    restaurant_id: '30112340'
  },
  {
    _id: ObjectId('6adb5f8adeafff9f132118bf'),
    address: {
      building: '7114',
      coord: [ -73.9068506, 40.6199034 ],
      street: 'Avenue U',
      zipcode: '11234'
    },
    borough: 'Brooklyn',
    cuisine: 'Delicatessen',
    grades: [
```

```
    grades: [
      { date: ISODate('2014-05-29T00:00:00.000Z'), grade: 'A', score: 10 },
      { date: ISODate('2014-01-14T00:00:00.000Z'), grade: 'A', score: 10 },
      { date: ISODate('2013-08-03T00:00:00.000Z'), grade: 'A', score: 8 },
      { date: ISODate('2012-07-18T00:00:00.000Z'), grade: 'A', score: 10 },
      { date: ISODate('2012-03-09T00:00:00.000Z'), grade: 'A', score: 13 },
      { date: ISODate('2011-10-14T00:00:00.000Z'), grade: 'A', score: 9 }
    ],
    name: "Wilken'S Fine Food",
    restaurant_id: '40356483'
  },
  {
    _id: ObjectId('6adb5f8adeafff9f132118c4'),
    address: {
      building: '1269',
      coord: [ -73.871194, 40.6730975 ],
      street: 'Sutter Avenue',
      zipcode: '11208'
    },
    borough: 'Brooklyn',
    cuisine: 'Chinese',
    grades: [
      { date: ISODate('2014-09-16T00:00:00.000Z'), grade: 'B', score: 21 },
      { date: ISODate('2013-08-28T00:00:00.000Z'), grade: 'A', score: 7 },
      { date: ISODate('2013-04-02T00:00:00.000Z'), grade: 'C', score: 56 },
      { date: ISODate('2012-08-15T00:00:00.000Z'), grade: 'B', score: 27 },
      { date: ISODate('2012-03-28T00:00:00.000Z'), grade: 'B', score: 27 }
    ],
    name: 'May May Kitchen',
    restaurant_id: '40358429'
  },
```

14. Find the average score for each restaurant.

```
24BYB0117> // Q14. Find the average score for each restaurant.
| db.restaurants.aggregate([
|   { $unwind: "$grades" },
|   {
|     $group: {
|       _id: "$restaurant_id",
|       name: { $first: "$name" },
|       averageScore: { $avg: "$grades.score" }
|     }
|   },
|   { $sort: { name: 1 } }
| ]);
[
  {
    _id: '40804423',
    name: '(Lewis Drug Store) Locanda Vini E Olii',
    averageScore: 6.6
  },
  {
    _id: '40359480',
    name: '1 East 66Th Street Kitchen',
    averageScore: 3.25
  },
  { _id: '40591271', name: '101 Deli', averageScore: 10.4 },
  { _id: '40560108', name: '101 Restaurant And Bar', averageScore: 10.6 },
  { _id: '40401104', name: '1020 Bar', averageScore: 18.333333333333332 },
  {
    _id: '40639261',
    name: '104-01 Foster Avenue Coffee Shop(Ups)',
    averageScore: 13
  },
  {
    _id: '40673117',
    name: '10Th Avenue Pizza & Cafe',
    averageScore: 9.5
  },
  { _id: '40884354', name: '111 Restaurant', averageScore: 9.75 },
  {
    _id: '40698892',
```

```
  {
    _id: '40639261',
    name: '104-01 Foster Avenue Coffee Shop(Ups)',
    averageScore: 13
  },
  {
    _id: '40673117',
    name: '10Th Avenue Pizza & Cafe',
    averageScore: 9.5
  },
  { _id: '40884354', name: '111 Restaurant', averageScore: 9.75 },
  {
    _id: '40698892',
    name: '15 East Restaurant',
    averageScore: 19.833333333333332
  },
  {
    _id: '40385555',
    name: '200 Fifth Avenue Restaurant & Sports Bar',
    averageScore: 19.833333333333332
  },
  { _id: '40364362', name: '21 Club', averageScore: 12 },
  { _id: '40388784', name: '2A', averageScore: 7.75 },
  {
    _id: '40592296',
    name: '3 Deli & Grill',
    averageScore: 9.666666666666666
  },
  { _id: '40391339', name: '3 Guys', averageScore: 12.333333333333334 },
  { _id: '40640495', name: '3 Guys Resturant', averageScore: 10.5 },
  { _id: '40605319', name: '42Nd Street Pizza Diner', averageScore: 8 },
  { _id: '40809948', name: "44 & X Hell'S Kitchen", averageScore: 9.5 },
  { _id: '40698807', name: '44 Sw Ristorante & Bar', averageScore: 13 },
  { _id: '40390163', name: '5 Burro Cafe', averageScore: 11 },
  {
    _id: '40806489',
    name: '525 Lex Restaurant & Bar',
```

15. Find the highest score for each restaurant.

```
24BYB0117> // Q15. Find the highest score for each restaurant.
| db.restaurants.aggregate([
|   { $unwind: "$grades" },
|   {
|     $group: {
|       _id: "$restaurant_id",
|       name: { $first: "$name" },
|       highestScore: { $max: "$grades.score" }
|     }
|   },
|   { $sort: { name: 1 } }
| ]);
[
  {
    _id: '40804423',
    name: '(Lewis Drug Store) Locanda Vini E Olii',
    highestScore: 12
  },
  {
    _id: '40359480',
    name: '1 East 66Th Street Kitchen',
    highestScore: 6
  },
  { _id: '40591271', name: '101 Deli', highestScore: 12 },
  { _id: '40560108', name: '101 Restaurant And Bar', highestScore: 15 },
  { _id: '40401104', name: '1020 Bar', highestScore: 38 },
  {
    _id: '40639261',
    name: '104-01 Foster Avenue Coffee Shop(Ups)',
    highestScore: 19
  },
  { _id: '40673117', name: '10Th Avenue Pizza & Cafe', highestScore: 12 },
  { _id: '40884354', name: '111 Restaurant', highestScore: 15 },
  { _id: '40698892', name: '15 East Restaurant', highestScore: 64 },
  {
    _id: '40385555',
    name: '200 Fifth Avenue Restaurant & Sports Bar',
    highestScore: 42
  },
  { _id: '40364362', name: '21 Club', highestScore: 12 },
  { _id: '40388784', name: '2A', highestScore: 12 },
  { _id: '40592296', name: '3 Deli & Grill', highestScore: 13 },
  { _id: '40391339', name: '3 Guys', highestScore: 26 },
  { _id: '40640495', name: '3 Guys Resturant', highestScore: 12 },
  { _id: '40605319', name: '42Nd Street Pizza Diner', highestScore: 10 },
  { _id: '40809948', name: "44 & X Hell'S Kitchen", highestScore: 13 },
  { _id: '40698807', name: '44 Sw Ristorante & Bar', highestScore: 19 },
  { _id: '40390163', name: '5 Burro Cafe', highestScore: 18 },
  { _id: '40806489', name: '525 Lex Restaurant & Bar', highestScore: 41 }
]
Type "it" for more
```

16. Find the lowest score for each restaurant.

```
24BYB0117> // Q16. Find the lowest score for each restaurant.
| db.restaurants.aggregate([
|   { $unwind: "$grades" },
|   {
|     $group: {
|       _id: "$restaurant_id",
|       name: { $first: "$name" },
|       lowestScore: { $min: "$grades.score" }
|     }
|   },
|   { $sort: { name: 1 } }
| ]);
[
  {
    _id: '40804423',
    name: '(Lewis Drug Store) Locanda Vini E Olii',
    lowestScore: 2
  },
  { _id: '40359480', name: '1 East 66Th Street Kitchen', lowestScore: 0 },
  { _id: '40591271', name: '101 Deli', lowestScore: 9 },
  { _id: '40560108', name: '101 Restaurant And Bar', lowestScore: 5 },
  { _id: '40401104', name: '1020 Bar', lowestScore: 7 },
  {
    _id: '40639261',
    name: '104-01 Foster Avenue Coffee Shop(Ups)',
    lowestScore: 8
  },
  { _id: '40673117', name: '10Th Avenue Pizza & Cafe', lowestScore: 7 },
  { _id: '40884354', name: '111 Restaurant', lowestScore: 3 },
  { _id: '40698892', name: '15 East Restaurant', lowestScore: 10 },
  {
    _id: '40385555',
    name: '200 Fifth Avenue Restaurant & Sports Bar',
    lowestScore: 12
  },
  { _id: '40364362', name: '21 Club', lowestScore: 12 },
  { _id: '40388784', name: '2A', lowestScore: 5 },
  { _id: '40592296', name: '3 Deli & Grill', lowestScore: 7 },
  { _id: '40391339', name: '3 Guys', lowestScore: 9 },
```

```
  { _id: '40592296', name: '3 Deli & Grill', lowestScore: 7 },
  { _id: '40391339', name: '3 Guys', lowestScore: 9 },
  { _id: '40640495', name: '3 Guys Resturant', lowestScore: 9 },
  { _id: '40605319', name: '42Nd Street Pizza Diner', lowestScore: 5 },
  { _id: '40809948', name: "44 & X Hell'S Kitchen", lowestScore: 3 },
  { _id: '40698807', name: '44 Sw Ristorante & Bar', lowestScore: 9 },
  { _id: '40390163', name: '5 Burro Cafe', lowestScore: 7 },
  { _id: '40806489', name: '525 Lex Restaurant & Bar', lowestScore: 7 }
]
Type "it" for more
24BYB0117>
```

17. Find the count of restaurants in each borough.

```
24BYB0117> // Q17. Find the count of restaurants in each borough.
| db.restaurants.aggregate([
|   { $group: { _id: "$borough", restaurantCount: { $sum: 1 } } },
|   { $sort: { _id: 1 } }
| ]);
[
  { _id: 'Bronx', restaurantCount: 309 },
  { _id: 'Brooklyn', restaurantCount: 684 },
  { _id: 'Manhattan', restaurantCount: 1883 },
  { _id: 'Queens', restaurantCount: 738 },
  { _id: 'Staten Island', restaurantCount: 158 }
]
```

18. Find the count of restaurants that received a grade of 'A' for each borough.

```
24BYB0117> // Q18. Find the count of restaurants that received a grade of 'A' for each borough.
| // This counts each restaurant once if it has at least one A grade.
| db.restaurants.aggregate([
|   { $match: { "grades.grade": "A" } },
|   { $group: { _id: "$borough", restaurantCount: { $sum: 1 } } },
|   { $sort: { _id: 1 } }
| ]);
[
  { _id: 'Bronx', restaurantCount: 308 },
  { _id: 'Brooklyn', restaurantCount: 683 },
  { _id: 'Manhattan', restaurantCount: 1877 },
  { _id: 'Queens', restaurantCount: 734 },
  { _id: 'Staten Island', restaurantCount: 157 }
]
```

19. Find the lowest score for each borough.

```
24BYB0117> // Q19. Find the lowest score for each borough.
| db.restaurants.aggregate([
|   { $unwind: "$grades" },
|   { $group: { _id: "$borough", lowestScore: { $min: "$grades.score" } } },
|   { $sort: { _id: 1 } }
| ]);
[
  { _id: 'Bronx', lowestScore: 0 },
  { _id: 'Brooklyn', lowestScore: 0 },
  { _id: 'Manhattan', lowestScore: -1 },
  { _id: 'Queens', lowestScore: 0 },
  { _id: 'Staten Island', lowestScore: -1 }
]
```

20. Find the name and address of the restaurants that received a grade of 'A' on a specific date.

```
24BYB0117> // Q20. Find the name and address of the restaurants that received a grade of 'A' on a specific date.
| db.restaurants.find(
|   {
|     grades: {
|       $elemMatch: {
|         grade: "A",
|         date: ISODate("2014-08-11T00:00:00Z")
|       }
|     }
|   },
|   { _id: 0, name: 1, address: 1 }
| );
[
  {
    address: {
      building: '225',
      coord: [ -73.96485799999999, 40.761899 ],
      street: 'East   60 Street',
      zipcode: '10022'
    },
    name: 'Serendipity 3'
  },
  {
    address: {
      building: '358',
      coord: [ -73.963506, 40.758273 ],
      street: 'East 57 Street',
      zipcode: '10022'
    },
    name: "Neary'S Pub"
  },
  {
    address: {
      building: '1133',
      coord: [ -73.958885, 40.7745559 ],
      street: 'Lexington Avenue',
      zipcode: '10075'
    },
    name: 'Don Filippo Restaurant'
```

```
    name: 'Don Filippo Restaurant'
  },
  {
    address: {
      building: '3707',
      coord: [ -73.8924962, 40.7485672 ],
      street: '73 Street',
      zipcode: '11372'
    },
    name: 'Ready Penny Inn'
  },
  {
    address: {
      building: '232',
      coord: [ -73.9723721, 40.7506241 ],
      street: 'East   43 Street',
      zipcode: '10017'
    },
    name: 'Pietros'
  },
  {
    address: {
      building: '6',
      coord: [ -73.97937, 40.754292 ],
      street: 'East   44 Street',
      zipcode: '10017'
    },
    name: 'The Cornell Center Of New York'
  },
  {
    address: {
      building: '324',
      coord: [ -73.9936506, 40.7475513 ],
      street: 'Seventh Avenue',
      zipcode: '10001'
    },
    name: "Mustang Sally'S Restaurant"
  },
```

21. Find the name and address of the restaurants that have the word 'coffee' in their name.

```
24BYB0117> // Q21. Find the name and address of the restaurants that have the word 'coffee' in their name.
| db.restaurants.find(
|   { name: /coffee/i },
|   { _id: 0, name: 1, address: 1 }
| );
[
  {
    address: {
      building: '26',
      coord: [ -73.9983, 40.715051 ],
      street: 'Pell Street',
      zipcode: '10013'
    },
    name: 'Mee Sum Coffee Shop'
  },
  {
    address: {
      building: '1191',
      coord: [ -73.8513114, 40.8316981 ],
      street: 'Castle Hill Avenue',
      zipcode: '10462'
    },
    name: "Lulu'S Coffee Shop"
  },
  {
    address: {
      building: '10001',
      coord: [ -74.03345, 40.612598 ],
      street: '4 Avenue',
      zipcode: '11209'
    },
    name: 'Narrows Coffee Shop'
  },
  {
    address: {
      building: '893',
      coord: [ -73.98557920000002, 40.7686093 ],
      street: '9 Avenue',
      zipcode: '10019'
```

22. Find the cuisine type that is most likely to receive a "C" grade.

```
24BYB0117> // Q22. Find the cuisine type that is most likely to receive a "C" grade.
| db.restaurants.aggregate([
|   { $unwind: "$grades" },
|   { $match: { "grades.grade": "C" } },
|   { $group: { _id: "$cuisine", cGradeCount: { $sum: 1 } } },
|   { $sort: { cGradeCount: -1 } },
|   { $limit: 1 }
| ]);
[ { _id: 'American ', cGradeCount: 172 } ]
```

23. Find the restaurants that achieved the highest total score.

```
24BYB0117> // Q23. Find the restaurants that achieved the highest total score.
| db.restaurants.aggregate([
|   {
|     $project: {
|       restaurant_id: 1,
|       name: 1,
|       totalScore: { $sum: "$grades.score" }
|     }
|   },
|   {
|     $setWindowFields: {
|       sortBy: { totalScore: -1 },
|       output: { highestTotal: { $max: "$totalScore", window: { documents: ["unbounded", "unbounded"] } } }
|     }
|   },
|   { $match: { $expr: { $eq: ["$totalScore", "$highestTotal"] } } },
|   { $project: { _id: 0, restaurant_id: 1, name: 1, totalScore: 1 } }
| ]);
[
  {
    name: 'Nanni Restaurant',
    restaurant_id: '40366157',
    totalScore: 225
  }
]
```

24. Find the top 5 restaurants with the highest average score for each cuisine type, along with their average scores.

```
24BYB0117> // Q24. Find the top 5 restaurants with the highest average score for each cuisine type, along with their average scores.
| db.restaurants.aggregate([
|   { $unwind: "$grades" },
|   {
|     $group: {
|       _id: { cuisine: "$cuisine", restaurant_id: "$restaurant_id" },
|       name: { $first: "$name" },
|       averageScore: { $avg: "$grades.score" }
|     }
|   },
|   { $sort: { "_id.cuisine": 1, averageScore: -1, name: 1 } },
|   {
|     $group: {
|       _id: "$_id.cuisine",
|       restaurants: {
|         $push: {
|           restaurant_id: "$_id.restaurant_id",
|           name: "$name",
|           averageScore: "$averageScore"
|         }
|       }
|     }
|   },
|   { $project: { _id: 1, topFive: { $slice: ["$restaurants", 5] } } },
|   { $sort: { _id: 1 } }
| ]);
[
  {
    _id: 'Afghan',
    topFive: [
      { restaurant_id: '40589545', name: 'Khyber Pass', averageScore: 12 },
      {
        restaurant_id: '40616799',
        name: 'Afghan Kebab House #1',
        averageScore: 11
      },
      {
        restaurant_id: '40552806',
        name: 'Afghan Kebab House',
        averageScore: 9.4
      },
```

```
      {
        restaurant_id: '40868400',
        name: 'Ariana Kebab House',
        averageScore: 5
      }
    ]
  },
  {
    _id: 'African',
    topFive: [
      {
        restaurant_id: '40832718',
        name: 'Ebe Ye Yie African Restaurant',
        averageScore: 11.2
      },
      {
        restaurant_id: '40684161',
        name: 'Madiba',
        averageScore: 10.833333333333334
      },
      {
        restaurant_id: '40798504',
        name: 'Africana Restaurant',
        averageScore: 9.8
      },
      {
        restaurant_id: '40368021',
        name: 'African Terrace',
        averageScore: 3.5
      }
    ]
  },
  {
    _id: 'American ',
    topFive: [
      {
        restaurant_id: '40756344',
        name: 'West 79Th Street Boat Basin Cafe',
        averageScore: 36
```

25. Find the top 5 restaurants in each borough with the highest number of "A" grades.

```
24BYB0117> // Q25. Find the top 5 restaurants in each borough with the highest number of "A" grades.
| db.restaurants.aggregate([
|   { $unwind: "$grades" },
|   { $match: { "grades.grade": "A" } },
|   {
|     $group: {
|       _id: { borough: "$borough", restaurant_id: "$restaurant_id" },
|       name: { $first: "$name" },
|       aGradeCount: { $sum: 1 }
|     }
|   },
|   { $sort: { "_id.borough": 1, aGradeCount: -1, name: 1 } },
|   {
|     $group: {
|       _id: "$_id.borough",
|       restaurants: {
|         $push: {
|           restaurant_id: "$_id.restaurant_id",
|           name: "$name",
|           aGradeCount: "$aGradeCount"
|         }
|       }
|     }
|   },
|   { $project: { _id: 1, topFive: { $slice: ["$restaurants", 5] } } },
|   { $sort: { _id: 1 } }
| ]);
[
  {
    _id: 'Bronx',
    topFive: [
      {
        restaurant_id: '40369087',
        name: 'Blue Bay Restaurant',
        aGradeCount: 7
      },
      {
        restaurant_id: '40863575',
        name: 'Broadway Pizza & Pasta',
        aGradeCount: 7
```

```
    _id: 'Bronx',
    topFive: [
      {
        restaurant_id: '40369087',
        name: 'Blue Bay Restaurant',
        aGradeCount: 7
      },
      {
        restaurant_id: '40863575',
        name: 'Broadway Pizza & Pasta',
        aGradeCount: 7
      },
      { restaurant_id: '40399778', name: "Mcdonald'S", aGradeCount: 7 },
      { restaurant_id: '40738028', name: "Ali'S Roti Shop", aGradeCount: 6 },
      {
        restaurant_id: '40560308',
        name: 'Caridad Restaurant',
        aGradeCount: 6
      }
    ]
  },
  {
    _id: 'Brooklyn',
    topFive: [
      { restaurant_id: '40864834', name: 'Almar', aGradeCount: 7 },
      {
        restaurant_id: '40513021',
        name: 'Blue Mountain Restaurant & Jerk Center',
        aGradeCount: 7
      },
      { restaurant_id: '40399340', name: 'Coffee Shop', aGradeCount: 7 },
      {
        restaurant_id: '40399986',
        name: 'Fifth Ave Cafe /Diner',
        aGradeCount: 7
      },
      {
        restaurant_id: '40388260',
        name: 'Amnon Kosher Pizza',
        aGradeCount: 6
      }
```